KOBI – KOMUNITI BIJAK

A smarter way to care for our neighbourhood

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Submission Date : 23 October 2026

2. EXECUTIVE SUMMARY

Identifying the right place

KoBi – Komuniti Bijak is a neighbourhood application with an interactive 2D map that helps residents report facility issues, track repair progress and discover community and recycling process. The app connects residents with Sime Darby Property (SDP) to ensure that facilities are well maintained and that community engagement is made easier.

Our target users are residents living in the township and SDP staff responsible for facility management and community programmes. KoBi addresses three key problems: difficulty in identifying the exact facility to report, lack of clear updates on repair progress and challenges in finding community and recycling activities. By providing a simple, visual and centralized platform, KoBi makes neighbourhood care more transparent, convenient and engaging for everyone.

3. PROBLEM STATEMENT

IDENTIFYING THE RIGHT PLACE

Residents may find it difficult to explain exactly which facility is faulty, especially when several look alike. On the map, they select the specific object before submitting a report.

Objective :

Make reporting easier with a clear facility location, a short description and an optional photo.

Research on citizen participation applications identifies reporting problems with urban infrastructure maintenance as a common use of these tools (Bastos et al., 2022).

KNOWING WHAT HAPPENS NEXT

After submitting a report, residents want to know whether it has been received, is being addressed or has been resolved. SDP staff also need organized information to act on.

Objective :

Keep progress updates, staff notes and evidence of completed repairs in one place.

MISSING LOCAL ACTIVITIES

Community and recycling announcements may be scattered across different channels. Residents need an easy way to find out what is happening, where it takes place and how to join.

Objective :

Bring announcements and the option to join or cancel participation into the same app.

Dursun et al. (2024) explain that recycling apps can address knowledge barriers by helping users recognise recyclable materials and understand appropriate recycling methods. This supports KoBi’s provision of accepted-material and collection information.

4. PROPOSED SOLUTION AND KEY FEATURES

KEY FEATURES

KoBi provides and easy-to-use platform with the following key features:

FEATURE	DESCRIPTION
Interactive 2D map	Explore the neighbourhood and locate facilities such as streetlights, parks and community halls.
Facility issue reporting	Report issues with a description, optional photo and exact location.
Repair status tracking	View status : Received In progress Resolved, with staff notes and photos.
Pixel worker animation	A visual worker appears near the facility during repairs to make the process clear and friendly.
Community activities	Browse announcements and view details (date, time, location, instructions).
Recycling events	Get information on accepted materials, collection points and event timing.
Join / cancel participation	Register or withdraw from activities directly in the app.

5. USER WORKFLOW AND SYSTEM DESIGN

RESIDENT WORKFLOW

1. Open the neighbourhood map and tap a facility (e.g: streetlight)
2. Submit a report with description and optional photo.
3. View the report in the resident’s own list.
4. Track the status as SDP updates the progress.
5. Discover and join community or recycling activities.

5.2 SDP STAFF WORKFLOW

1. View new reports in the staff dashboard.
2. Update the status: Received □ In progress □ Resolved.
3. Add staff notes and photos as evidence.
4. The update is visible to the resident on the map in their list.

6. TECHNICAL IMPLEMENTATION

TECHNOLOGIES USED

The project uses the following technologies:

COMPONENT	TECHNOLOGY
Programming and markup	JavaScript (ES modules), JSX, HTML5 and plain CSS for application logic, page structure and styling.
Frontend UI library	React 19.3.1 and React DOM 19.3.1 for reusable components and interactive resident and SDP screens.
Development and build tools	Vite 8.3.4 with the React plugin for the development server and production builds.
Testing and code quality	Vitest for unit tests, Playwright for browser tests and ESLint for code checks.
Planned 2D world framework	Phaser 3.90.0 for the pixel-art map and world interactions; planned and not yet installed in the current frontend shell.

7. EXPECTED IMPACT AND EVALUATION

7.1 EXPECTED BENEFITS

STAKEHOLDER	BENEFITS
Residents	Easier reporting, clearer progress updates and simpler activity discovery.
SDP	More organized reports and efficient facility management.
Community	Stronger engagement and a cleaner, safer neighbourhood.
Environment	More recycling participation and reduced waste.

7.2 EVALUATION METRICS (PROPOSED)

Time taken to submit a report

Completeness of submitted reports

Time taken to handle issues

Number of activity registrations

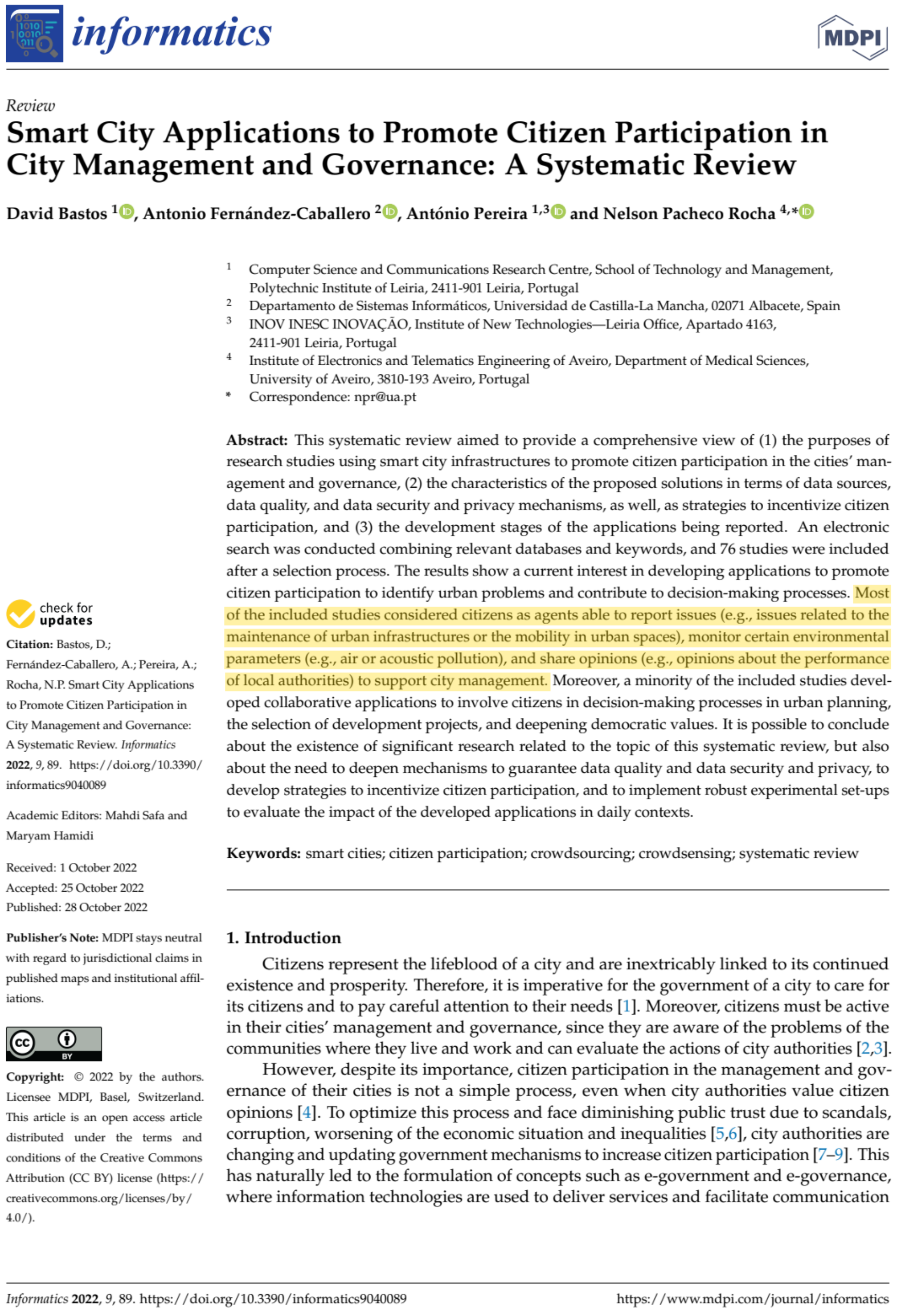
8. REFERENCES

Bastos, D., Fernández-Caballero, A., Pereira, A., & Rocha, N. P. (2022). Smart city applications to promote citizen participation in city management and governance: A systematic review. *Informatics, 9*(4), Article 89. https://doi.org/10.3390/informatics9040089

Dursun, İ., Tümer, E., & Yürüyen Kılıç, H. (2024). Roles of mobile applications in removing barriers to individual recycling: Case of Türkiye. *Environmental Research and Technology, 7*(1), 97–107. https://doi.org/10.35208/ert.1317616

APPENDIX A — CITATION EVIDENCE

Bastos et al. (2022), article page 1 (PDF page 2). The highlighted abstract passage supports the infrastructure-reporting statement in Section 3.



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Review

Smart City Applications to Promote Citizen Participation in City Management and Governance: A Systematic Review

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Abstract: This systematic review aimed to provide a comprehensive view of (1) the purposes of research studies using smart city infrastructures to promote citizen participation in the cities' management and governance, (2) the characteristics of the proposed solutions in terms of data sources, data quality, and data security and privacy mechanisms, as well, as strategies to incentivize citizen participation, and (3) the development stages of the applications being reported. An electronic search was conducted combining relevant databases and keywords, and 76 studies were included after a selection process. The results show a current interest in developing applications to promote citizen participation to identify urban problems and contribute to decision-making processes. Most of the included studies considered citizens as agents able to report issues (e.g., issues related to the maintenance of urban infrastructures or the mobility in urban spaces), monitor certain environmental parameters (e.g., air or acoustic pollution), and share opinions (e.g., opinions about the performance of local authorities) to support city management. Moreover, a minority of the included studies developed collaborative applications to involve citizens in decision-making processes in urban planning, the selection of development projects, and deepening democratic values. It is possible to conclude about the existence of significant research related to the topic of this systematic review, but also about the need to deepen mechanisms to guarantee data quality and data security and privacy, to develop strategies to incentivize citizen participation, and to implement robust experimental set-ups to evaluate the impact of the developed applications in daily contexts.

Keywords: smart cities; citizen participation; crowdsourcing; crowdsensing; systematic review

1. Introduction

Citizens represent the lifeblood of a city and are inextricably linked to its continued existence and prosperity. Therefore, it is imperative for the government of a city to care for its citizens and to pay careful attention to their needs [1]. Moreover, citizens must be active in their cities' management and governance, since they are aware of the problems of the communities where they live and work and can evaluate the actions of city authorities [2,3].

However, despite its importance, citizen participation in the management and governance of their cities is not a simple process, even when city authorities value citizen opinions [4]. To optimize this process and face diminishing public trust due to scandals, corruption, worsening of the economic situation and inequalities [5,6], city authorities are changing and updating government mechanisms to increase citizen participation [7–9]. This has naturally led to the formulation of concepts such as e-government and e-governance, where information technologies are used to deliver services and facilitate communication

check for updates

Citation: Bastos, D.; Fernández-Caballero, A.; Pereira, A.; Rocha, N.P. Smart City Applications to Promote Citizen Participation in City Management and Governance: A Systematic Review. *Informatics* **2022**, *9*, 89. https://doi.org/10.3390/informatics9040089

Academic Editors: Mahdi Safa and Maryam Hamidi

Received: 1 October 2022
Accepted: 25 October 2022
Published: 28 October 2022

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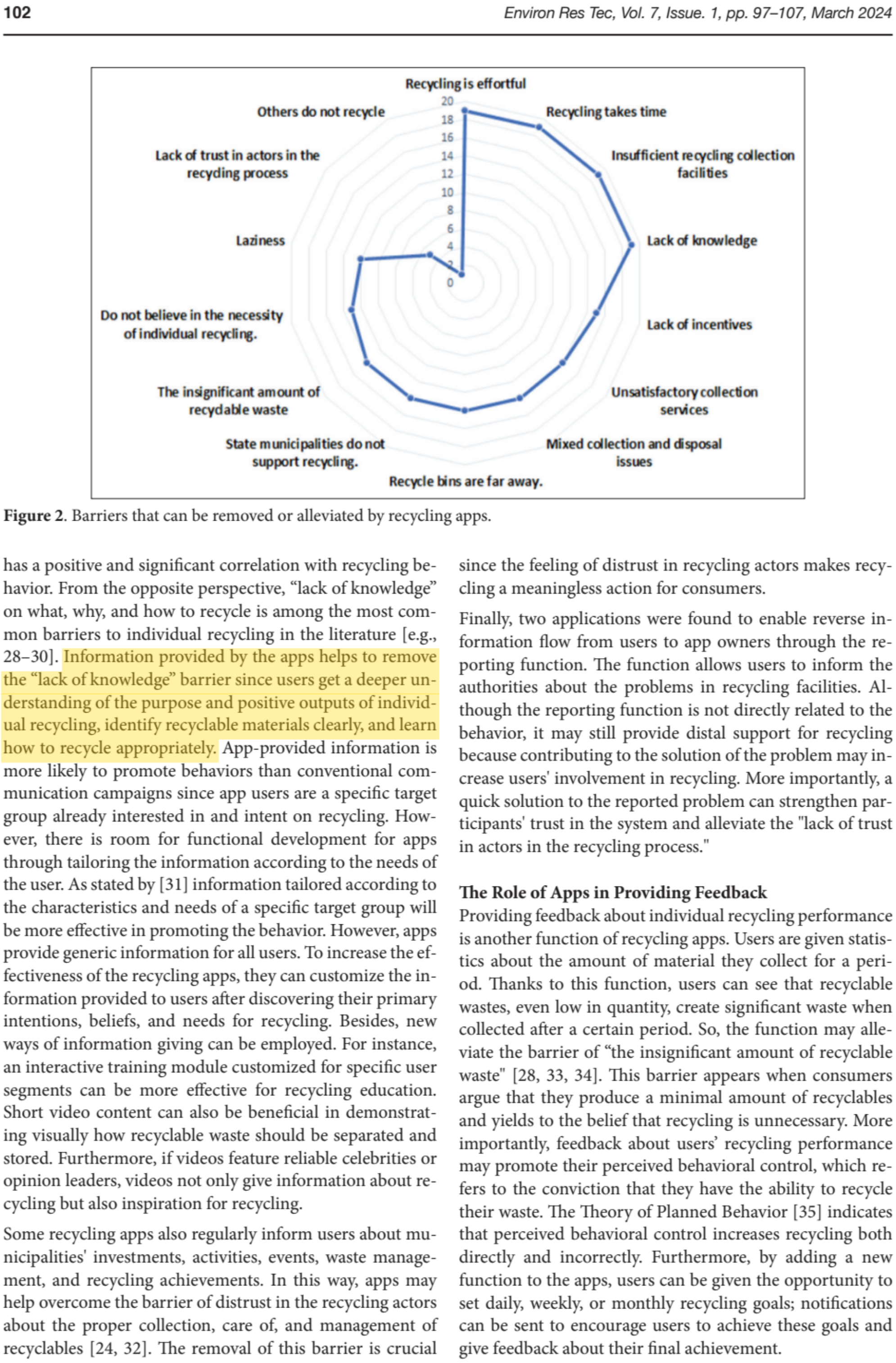
Informatics **2022**, *9*, 89. https://doi.org/10.3390/informatics9040089 https://www.mdpi.com/journal/informatics

APPENDIX B — CITATION EVIDENCE

Dursun et al. (2024), article page 102 (PDF page 6). The highlighted passage supports the recycling-information statement in Section 3.



Recycling is effortful
Recycling takes time
Insufficient recycling collection facilities
Lack of knowledge
Lack of incentives
Unsatisfactory collection services
Mixed collection and disposal issues
Recycle bins are far away.
State municipalities do not support recycling.
The insignificant amount of recydable waste
Do not believe in the necessity of individual recycling.
Laziness
Lack of trust in actors in the recyding process
Others do not recycle
20
18
16
14
12
10
8
6
4
2
0

Figure 2. Barriers that can be removed or alleviated by recycling apps.

has a positive and significant correlation with recycling behavior. From the opposite perspective, "lack of knowledge" on what, why, and how to recycle is among the most common barriers to individual recycling in the literature [e.g., 28–30]. Information provided by the apps helps to remove the "lack of knowledge" barrier since users get a deeper understanding of the purpose and positive outputs of individual recycling, identify recyclable materials clearly, and learn how to recycle appropriately. App-provided information is more likely to promote behaviors than conventional communication campaigns since app users are a specific target group already interested in and intent on recycling. However, there is room for functional development for apps through tailoring the information according to the needs of the user. As stated by [31] information tailored according to the characteristics and needs of a specific target group will be more effective in promoting the behavior. However, apps provide generic information for all users. To increase the effectiveness of the recycling apps, they can customize the information provided to users after discovering their primary intentions, beliefs, and needs for recycling. Besides, new ways of information giving can be employed. For instance, an interactive training module customized for specific user segments can be more effective for recycling education. Short video content can also be beneficial in demonstrating visually how recyclable waste should be separated and stored. Furthermore, if videos feature reliable celebrities or opinion leaders, videos not only give information about recycling but also inspiration for recycling.

Some recycling apps also regularly inform users about municipalities' investments, activities, events, waste management, and recycling achievements. In this way, apps may help overcome the barrier of distrust in the recycling actors about the proper collection, care of, and management of recyclables [24, 32]. The removal of this barrier is crucial since the feeling of distrust in recycling actors makes recycling a meaningless action for consumers.

Finally, two applications were found to enable reverse information flow from users to app owners through the reporting function. The function allows users to inform the authorities about the problems in recycling facilities. Although the reporting function is not directly related to the behavior, it may still provide distal support for recycling because contributing to the solution of the problem may increase users' involvement in recycling. More importantly, a quick solution to the reported problem can strengthen participants' trust in the system and alleviate the "lack of trust in actors in the recycling process."

The Role of Apps in Providing Feedback

Providing feedback about individual recycling performance is another function of recycling apps. Users are given statistics about the amount of material they collect for a period. Thanks to this function, users can see that recyclable wastes, even low in quantity, create significant waste when collected after a certain period. So, the function may alleviate the barrier of "the insignificant amount of recyclable waste" [28, 33, 34]. This barrier appears when consumers argue that they produce a minimal amount of recyclables and yields to the belief that recycling is unnecessary. More importantly, feedback about users' recycling performance may promote their perceived behavioral control, which refers to the conviction that they have the ability to recycle their waste. The Theory of Planned Behavior [35] indicates that perceived behavioral control increases recycling both directly and incorrectly. Furthermore, by adding a new function to the apps, users can be given the opportunity to set daily, weekly, or monthly recycling goals; notifications can be sent to encourage users to achieve these goals and give feedback about their final achievement.